Analysis of Skin Color Models for Face Detection

Mrunmayee V. Daithankar, Kailash J. Karande, Avinash D. Harale

***Abstract*— this paper describes face detection method using skin color information. Processing color information is faster than processing other facial features. There are different color models present for skin color segmentation like RGB, CMY, YUV, YIQ, YCbCr, YCgCr, HSI, etc. But to choose best option among these models for efficient skin color segmentation is difficult task. In this paper we have explored and analyse three most popular color models-YCbCr, YCgCr and HSI for face detection.**

***Index Terms*— Face detection, skin color segmentation, color models- YCbCr, YCgCr and HSI, face localization.**

I. INTRODUCTION

Face detection means to check whether the face is present in image or not. In the face localization process we can determine the exact location of face in image. Many researchers work in this area because; this is initial and very important step in many applications such as Facial Expression Analysis, Human Computer Interface (HCI), Security Systems, Face Recognition, Surveillance Systems, Image Enhancement, etc.

Different methods by which we can detect faces are categorised as follows [1]:

-Template matching methods,
-Knowledge based methods,
-Machine learning based methods and
-Feature based methods

Every method has its own advantages and disadvantages. If we choose feature based methods, it is more advantageous for us, because these are scale independent, rotation independent, and fast. The features comes under this technique are color, edge, shape and texture to detect faces [2]. Many authors found that color processing is faster than processing other facial features. The skin color detection plays important role in face detection. There are different skin color models used for skin color segmentation such as, RGB, YIQ, YUV, CIELab, YCbCr, YCgCr, HSI, etc. [3]. But to choose best model among these, whose performance is better in skin color segmentation is difficult task.

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In this paper we are going to study and analyse three most popular color models-YCbCr, YCgCr and HSI.

A. Skin Color Models:

Skin segmentation using different skin color models for face detection has many advantages as this technique is scale independent, rotation independent, and fast. By using this method we can reduce computational cost for face detection. Different color models used for skin color segmentation are, RGB, YIQ, YUV, CIELab, YCbCr, YCgCr, HSI, etc. [3].

Among these we are choosing YCbCr, YCgCr and HSI color models which are most popular for efficient face detection. Overview of these color models is as follows:

1) YCbCr Color Model:

The YCbCr color model represents image in three components; one is luminous component and remaining two are chrominance components [4]. The chrominance components are not depends on luminance component. In this color model, Y - Luminance component i.e. light intensity,
Cb- Blue difference of the chromaticity component and
Cr- Red difference of the chromaticity component

The relationship between chrominance and luminance of the skin color is non-linear in the high and low luminance region. The transformation between RGB and YCbCr model is represented by the following equations:

$$Y = 16 + 1/256 \times (65.738R + 129.057G + 25.064B) \quad (1)$$
$$Cb = 128 + 1/256 \times (-37.945R - 74.494G + 112.439B) \quad (2)$$
$$Cr = 128 + 1/256 \times (112.439R - 94.154G - 18.285B) \quad (3)$$

The range in which Y, Cb and Cr components are scaled is given below:
Y=16 to 235, where 16 for black and 235 for white. Cb=16 to 240 and Cr =16 to 240 [4].

As the luminous and chrominance components can be treated separately, the influence of illumination can be removed during processing of an image [5]. This is the main advantage of YCbCr color model.

2) YCgCr Color Model:

This color space also represents one luminance component and two chrominance components like YCbCr color model. The only difference between these two color models is, YCbCr model uses the Cb component for chrominance, while YCgCr uses the Cg component for the second channel. In the YCgCr color space, a human skin color model can be concentrated in

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a small region of the Cg-Cr plane [6], [7]. This color space includes information about green difference instead of blue difference, which can be more useful for skin color detection. The conversion between RGB and this color model is represented by following equations:

$$Y = 16 + 65.481R + 128.553G + 24.966B \quad (4)$$
$$Cg = 128 - 81.085 + 112G - 30.915B \quad (5)$$
$$Cr = 128 + 112R - 93.768G - 18.214B \quad (6)$$

Where, range of R, G and B components is from 0 to 1; and Y, Cg and Cr components are scaled in the range of 16 to 240.

3) HSI Color Model:

In this color model, H-Hue describes the main color i.e. depth of color and ranges from 0 to 360. S-saturation gives purity of the color and its range is from 0 to 100 %. If the saturation is low then more "grayness" of color is represented. I-Intensity indicates the brightness of the shade. HSI color model has been used for image processing because it can separate the chromaticity from the intensity of the image.

If saturation is low then, Hue (H) is not reliable for the discrimination task. But for color description, HSI color model is preferred over RGB color model [8]. The conversion between RGB to HSI color model is as shown in following equations:

$$I = (R + G + B) / 3 \quad (7)$$

$$S = 1 - [3 \times (\min(R, G, B) / (R + G + B))] \quad (8)$$

$$\theta = [(R-G) + (R-B)] / [2\times [(R-G)\hat{}2 + (R-B)(G-B)] \hat{}0.5] \quad (9)$$

$$\text{If } G \geq B \text{ then } H = \theta \text{ otherwise } H = 360 - \theta \quad (10)$$

In this paper, Section I. Introduces to skin color models. Section II. Gives related work on face detection. Section III. Provide details of proposed work. Section IV. Gives experimental results carried out with proposed work. Section V. This section concludes the paper. And last part of paper contains references.

II. Related Work

There are different theories for face detection and localization using skin color segmentation presented by different authors.

A new method of face detection using combined skin color detector and template matching method is explored by Smita Tripathi et al. [9]. The template matching method is used to remove non-faces and to detect faces more accurately. An improved skin segmentation algorithm for face detection in color images with RGB-HS-CbCr color model is proposed by Sayantan Thakur et al. [10].

Nidhi Tiwari et al. [11] describes new method for face detection which combines skin color model, edge detection and face verification. In this paper, to locate mouth, eyes and nose positions, a novel technique of facial geometry is used. The study of different color models like RGB, YCbCr, HSV and CIELAB is done in the paper of Ekta Rewar et al. This paper compares these color models based on detection rate of skin regions and the results shows that YCbCr color model yields best output [12]. An effective face detection algorithm based on skin color information and neural network is presented by Alaa Sagheer et al. [13].

Khalid Mohamed Alajel et al. [14] proposed a new face detection algorithm by using skin color modeling and modified Hausdorff distance. The technique of fusion of multi color space for human skin region segmentation is given by Fan Hai Xiang et al. [15].

III. Proposed Work

The proposed system for face detection is based on color information, so the input image must be color image. The flow of proposed work is as shown in Fig.1. This system will not work with gray scale images. The input image may be of any kind so to get better output, enhance the input image. This step may contain any operation like morphological filtering, noise removal, image conversion etc. as per requirement.

In the step of feature extraction, we are using "color" as unique feature. This feature is scale-independent, rotation independent and fast which can reduce computational cost of face detection. For skin tone segmentation different color models like YCbCr, YCgCr and HSI are used. The simplest algorithm for color segmentation is selection of threshold for each channel of color models. For color model conversion different equations are present for different color models. To convert RGB image in YCbCr color model we have used eq. (1), (2) and (3). To get YCgCr color image we have used eq. (4), (5) and (6).Similarly, to convert RGB image in HSI image we have eq. (7), (8), (9) and (10).

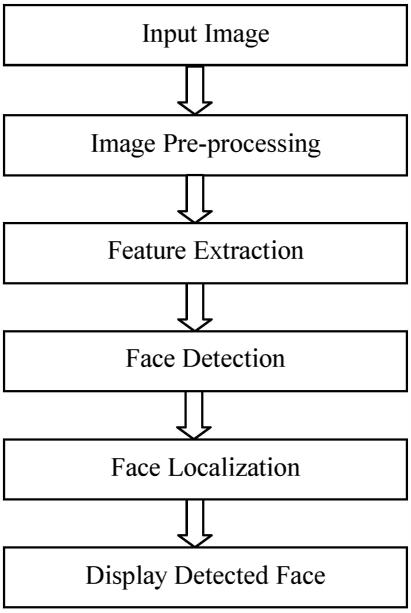

Fig.1.Outline of proposed work

After color model conversion, image is segmented in skin and non-skin region. This skin image may contain noise or such pixels which are like skin pixels but actually they are not. To get smooth skin region we can apply morphological operations like erosion or dilation or both. By doing this we can reach very close to our area of interest. The segmented skin region may contain skin of any body part, so to avoid confusion we are using height to width ratio which is in the specific range for face region.

IV. Experimental Results

The experimentation is carried out with the help of MATLAB software. To analyze performance of different color models in the skin segmentation we have given some images as input to three most popular color models. The color models used for skin segmentation are YCbCr, YCgCr and HSI. The performance of these three models is shown in following steps with figures:

1) Input image: Take RGB image as input to system. We have taken different images with different backgrounds for experimentation. One of the input images is as shown in Fig.2.

2) Skin Color Segmentation: To segment the skin region, input image is converted into skin color models. The segmentation is done by choosing threshold values for each channel of color model. The skin segmented images by YCbCr color model is as shown in Fig.3.a. Similarly the skin segmented image by YCgCr and HSI color model is as shown in Fig.3.b. and Fig.3.c. respectively.

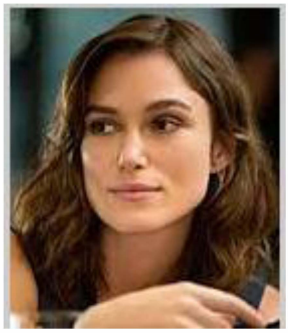

Fig.2. RGB Input Image

Fig.3.a. YCbCr-Skin Segmentation

Fig.3.b. YCgCr-Skin Segmentation

Fig.3.c. HSI-Skin Segmentation

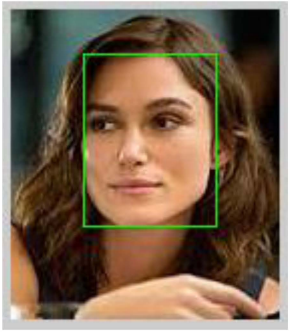

Fig.4.a.YCbCr result

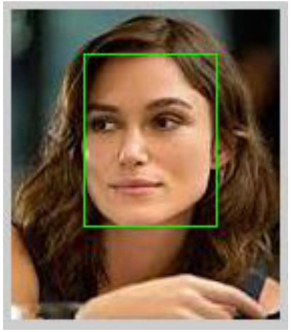

Fig.4.b.YCgCr result

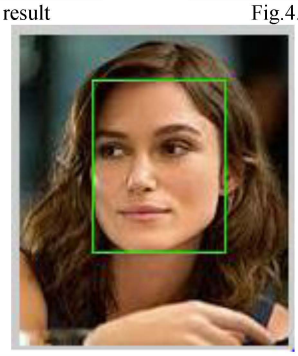

Fig.4.c. HSI result

3) Face Detection: The skin segmented image may contain any body part like hand or neck. By using height to width ratio we can detect only face region. Thus detected face images by using three different color models are shown in Fig 4.a., Fig 4.b. and Fig 4.c. The results of experimentation in percentage with different color models are given in TABLE I.

The purpose of study these color models is to integrate uncommon features of all models for efficient face detection [16]. This integrated model which is combination of YCbCr, YCgCr and HSI color model is as shown in Fig.5. More face detection results are shown in Fig. 6.

TABLE I
FACE DETECTION RATE OF DIFFERENT COLOR MODELS

Color Models	Face Detection Rate
1] YCbCr Color Model	93.75%
2] YCgCr Color Model	75%
3] HSI Color Model	87.5%

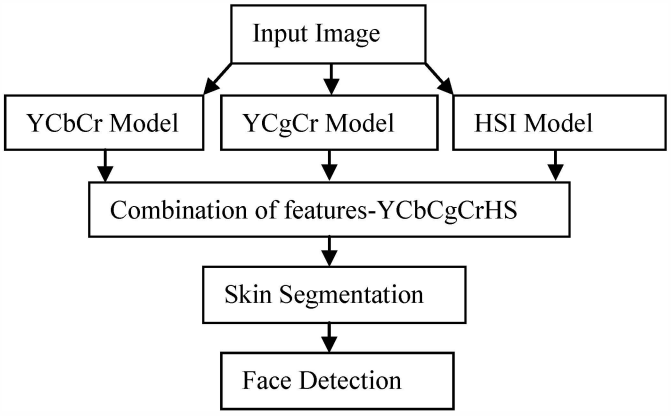

Fig.5. Integrated model for face detection

V. CONCLUSION

In this paper, skin color detection algorithm is used for face detection which is scale and rotation independent. For skin color segmentation different color models are present. We have done analysis of three most popular color models- YCbCr, YCgCr and HSI. Among these, YCbCr color model shows better results than others. HSI model also shows good results. The proposed work is simple and gives efficient face detection rate among others discussed in related work.

In future we want to explore the integrated model which may give us better results for efficient face detection as compare to any single color model.

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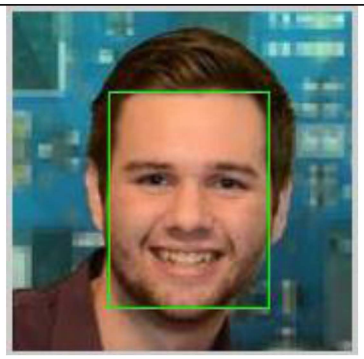
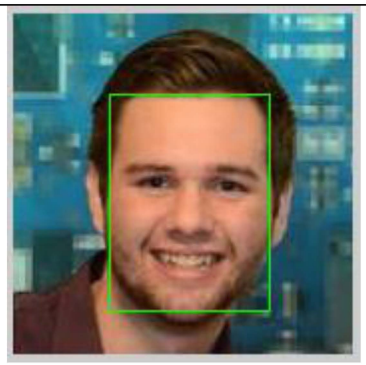
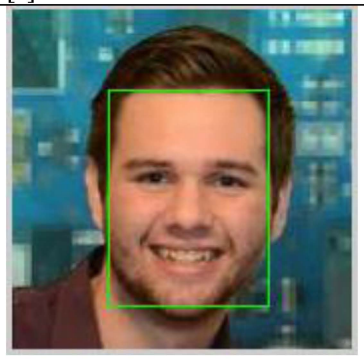
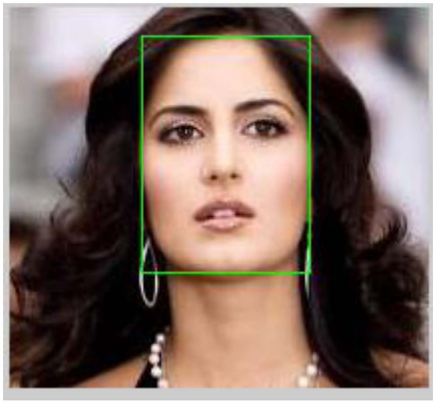
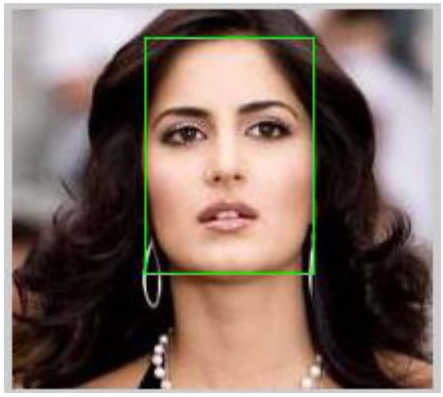
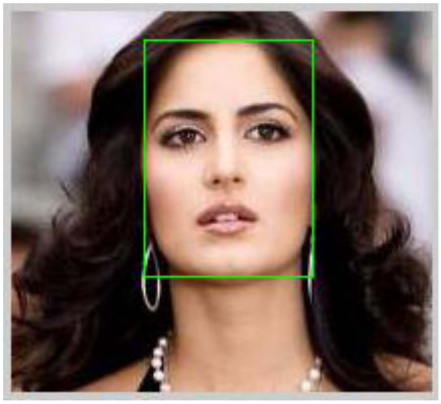
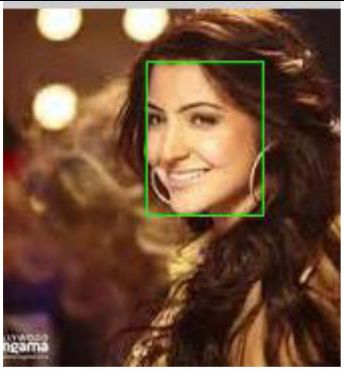
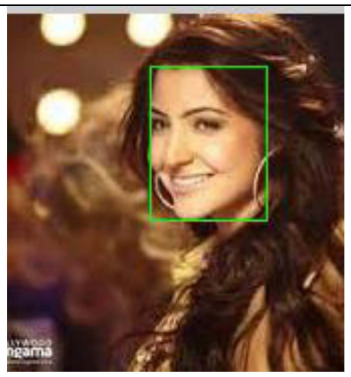
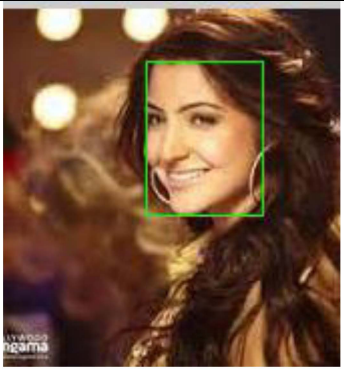
More results for face detection			
Sr. No.	YCbCr Color Model	HSI Color Model	YCgCr Color Model
1	[a]	[b]	[c]
2	 [a]	 [b]	 [c]
3	 [a]	 [b]	 [c]
4	 [a]	 [b]	 [c]

Fig.6. More face detection results with different skin color models